

# GCS Mission Planner — Implementation Report

## Executive Summary

This Flutter GCS (Mission Planner) implementation is a **map-first tactical UI shell** with sidebar navigation, top status bar, right telemetry panel, and a bottom mission dock. The UI has been upgraded from “demo mission planner” to a **flight-ready GCS interface (UI only)**: flight status, link quality, alerts, pre-arm checks, manual control panel, map HUD overlays, home marker, flight trail, RTH feedback, and mission progress. **Real vehicle integration (MAVLink), calibration execution, mission upload, and hardware control are not implemented** yet—vehicle-critical actions remain UI/demo placeholders, but the UI is structured to wire to MAVLink later.

## 1) Current Scope & Runtime Environment

- **Platform:** Flutter Web (Chrome) in this workspace
- **UI Style:** Dark tactical GCS theme (panel borders, glow accents, frosted top bar)
- **Data Source:** **Simulated telemetry** + UI-only flight/mission state
- **Real FC Link:** **Not present** (no MAVLink parsing/transport)

## 2) Application Layout & Main Components

### 2.1 Top Status Bar (Header Strip)

#### Purpose

- Shows flight-critical indicators and key status:
- Link (OK/LOST) + signal %
- Latency + link type (WiFi/Radio/USB)
- GPS summary
- Battery summary
- Flight mode + system state (IDLE / FLYING / LANDING / EMERGENCY)
- Arm state (color-coded)

#### Controls

- Tools button (opens Tools sheet)
- Telemetry toggle (opens/closes right telemetry panel)
- Close page (returns to map when on overlay pages)

#### Works properly

- Status text updates live (from the shared status model driven by the simulator)
- Tools sheet opens reliably
- Telemetry panel toggle works (open/close)
- Color-coded state/mode pill updates based on armed + mode

#### Not real yet

- Link/connection status is not backed by any real transport or vehicle connection

### 2.2 Left Sidebar (Navigation Hub)

#### Sections

- Map
- Mission

- Logs
- Settings
- Tools

#### **Works properly**

- Selected item highlight
- Collapse/expand via chevron handle
- Tools opens tool sheet
- Navigation switches content sections

#### **Not real yet**

- Some pages are UI scaffolds (depends on route/page implementation)

### **2.3 Center Map (Primary Area)**

#### **Purpose**

- Primary operational surface: basemap, overlays, waypoints, mission geometry, vehicle marker

#### **Works properly**

- Map renders (`flutter\_map`) and responds to interactions
- Basemap switching (dark/satellite) works
- Airspace overlay toggle works (demo NFZ logic)
- Mission geometry toggle works (route + waypoint markers)
- Vehicle marker + heading updates smoothly (simulated)
- Follow/tracking updates camera position (throttled for web performance)
- Mode-based feedback:
- Temporary mode banner
- SnackBar "Switched to "
- Zoom rail:
- Works
- Can be positioned to track the right telemetry strip (GCS shell integration)
- Map HUD overlay (toggleable):
- Compass/heading chip + home direction arrow
- Speed and altitude
- Distance to home
- Home marker on map:
- "HOME" marker always visible
- Displays distance to home
- Flight path trail (toggleable):
- Polyline track of recent vehicle movement

#### **Not real yet**

- No MAVLink vehicle position/attitude
- No real airspace feeds; NFZ is demo logic

### **2.4 Right Telemetry Panel (Collapsible Overlay)**

#### **Purpose**

- Vertical telemetry deck (cards) with flight-critical metrics:
- Altitude (MSL)
- Speed
- Climb rate
- Battery %

- Voltage (V), Current (A), Time remaining estimate
- GNSS sats + HDOP
- Fix type + accuracy estimate (UI-only)
- EKF / IMU / Compass health (UI-only)
- Heading

#### **Works properly**

- Collapsible handle works (panel open/closed)
- Width clamps to avoid overflow
- Telemetry values update in real time (simulated)

#### **Not real yet**

- Not backed by MAVLink messages

### **2.5 Bottom Dock (Mission Controls + Logs + Graph Stub)**

#### **States**

- Compact: toolbar row
- Expanded: toolbar + logs + graph stub

#### **Toolbar buttons present**

- Start Mission
- Load
- Save
- Pause (placeholder)
- RTH (UI mode switch placeholder)
- Waypoints (opens waypoint property sheet)
- Summary (opens mission summary sheet)
- Track (toggles follow/tracking)
- KILL / EMERGENCY STOP (UI-only; long-press + optional confirm dialog)

#### **Works properly**

- Expand/collapse dock works
- Horizontal toolbar scroll prevents overflow
- Buttons are sized tighter to keep fit
- Logs list renders (demo lines)
- Graph area renders stub “ALT live graph”

#### **Not real yet**

- Start/Pause/RTH/E-Stop do not send commands to a vehicle
- Load/Save are not a full mission file pipeline to/from FC (currently demo dialogs)

## **3) Mission / Waypoint Features**

### **3.1 Waypoint Creation**

#### **Current UX**

- Long-press map to add waypoint (demo behavior)
- “Add waypoint” appears in map actions menu

#### **Works properly**

- Waypoint list/route visuals are shown when mission geometry is enabled
- Demo NFZ violation blocks mission start (UI gating)

#### **Not real yet**

- No upload/download via MAVLink mission protocol
- No persistence to vehicle/autopilot

### **3.2 Mission Summary & Start Gating**

#### **Summary sheet shows**

- Waypoint count
- Pre-arm state (demo)
- Airspace violation state (demo)
- Ready/Blocked

#### **Works properly**

- Ready/Blocked updates from demo conditions (NFZ + pre-arm)
- Waypoint editor opens (property sheet)
- Pre-arm checklist is “dynamic-ready”:
- Checklist items can be initialized from live state (GPS lock, battery, EKF, link) while still allowing UI override in demo mode

#### **Not real yet**

- No true pre-arm checks from autopilot
- No mission start/monitor via real vehicle

### **3.3 Map Actions Menu (“More actions”)**

#### **Includes**

- Add waypoint
- Map layers
- Arm/Disarm (demo)
- Flight mode select (dialog)
- New mission wizard
- Survey grid (placeholder)
- Import mission (placeholder)

#### **Works properly**

- Menu and dialogs behave correctly
- Mode selection updates UI mode state
- Arm/disarm toggles simulated armed state
- Map layers sheet toggles overlays and UI panels:
- Airspace & NFZ
- Mission route & waypoints
- Vehicle marker
- Flight path trail
- Map HUD
- Manual control panel

#### **Not real yet**

- No arm/disarm/mode commands sent to autopilot

#### **4) Tools Sheet (Secondary Panels / Utilities)**

##### **Access**

- From top bar Tools icon
- From sidebar Tools item

##### **Entries**

- Telemetry
- Camera
- Manual control
- Airspace
- Replay
- Plugins
- Vehicles
- Vehicle setup

##### **Works properly**

- Sheet opens and navigates to routes

##### **Not real yet**

- Most tools/routes are UI scaffolding unless wired to real data/services

#### **5) Flight-Critical UI Additions (UI-only, Dynamic-ready)**

##### **5.1 Flight status indicator**

- Arm state (color-coded)
- Mode display
- System state (IDLE / FLYING / LANDING / EMERGENCY)

##### **5.2 Connection & signal indicator**

- Signal strength % (UI-only)
- Latency (ms)
- Link type label (WiFi/Radio/USB)

##### **5.3 Global alert / warning system**

- Central alert store (`GcsStatusModel`)
- Banner alert surface (critical/persistent)
- Alert types supported (UI-only triggers):
- LOW BATTERY
- GPS LOST
- SIGNAL LOST / Link degraded
- EKF ERROR (heuristic)
- COMPASS ERROR (heuristic)

##### **5.4 Pre-arm checklist panel**

Checklist items:

- GPS lock
- Battery OK
- Sensors calibrated (UI proxy)

- EKF healthy (UI proxy)
- RC signal OK (UI proxy)

### **5.5 Manual control panel**

- Collapsible panel overlay (sliders)
- Shows live values for Throttle/Yaw/Pitch/Roll (UI-only)
- Positioned to preserve map visibility and avoid the bottom dock

### **5.6 Map HUD overlay**

- Compass / heading indicator
- Home direction arrow
- Speed and altitude
- Distance to home

### **5.7 Flight path trail**

- Polyline trail of recent simulated movement
- Toggle on/off

### **5.8 RTH feedback**

- When RTH is active (mode = RTH + armed):
- “Returning to Home”
- Distance remaining
- Phase (Travel / Descend / Land)

### **5.9 Mission progress UI**

- When mission is active (UI-only condition):
- Current waypoint index
- Distance to next waypoint
- Completion %

## **6) Responsiveness / Non-overflow Handling (Current Quality)**

### **Solid behaviors**

- Sidebar collapse/expand transitions are smooth
- Telemetry panel width clamps to prevent overflow
- Dock toolbar scroll prevents RenderFlex overflow
- Map update cadence throttled for web performance

### **Known limitation**

- This is still an overlay-heavy shell (map + panels). It is stable visually, but not yet a fully “native” non-overlay grid layout.

## **7) What's Missing for “Real GCS” Operation (Critical Gaps)**

### **6.1 Vehicle Link (Required)**

- MAVLink transport: USB serial / telemetry radio / UDP
- MAVLink parsing and message routing
- Command/ack handling

- Heartbeat and link health computation

## 6.2 Calibration (Required for real setup)

- Sensor calibration flows via MAVLink commands
- Progress / instruction feedback (STATUSTEXT, ACKs)
- Parameter access (read/write, persistence, reboot workflows)

## 6.3 Mission Protocol (Required)

- Upload/download mission items
- Start/stop/monitor mission
- Geofence/failsafe handling

## 6.4 Safety & Reliability (Strongly recommended)

- State gating (disallow calibration while armed)
- Failsafe surfaces (link loss, EKF errors, GPS loss)
- Proper emergency stop behavior (kill/disarm) where supported

## 8) Feature Status Matrix (■ / ■ / ■)

|      |         |        |                   |       |                          |   |  |       |                              |            |  |       |                             |            |  |       |                               |   |  |     |                    |   |  |     |                           |         |  |     |           |   |  |     |                 |            |  |     |             |        |  |     |              |            |  |     |                      |        |  |     |                     |            |  |     |                         |            |  |         |                          |          |  |         |                           |          |  |         |                         |   |  |           |                            |         |  |           |                                        |            |  |      |                             |          |  |      |                          |        |  |  |          |                 |                  |  |  |             |  |                      |   |  |  |              |                   |   |  |
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| Area | Feature | Status | ----- ----- ----- | Shell | Top status bar + actions | ■ |  | Shell | Flight status + link/latency | ■ (sim/UI) |  | Shell | Global alerts banner system | ■ (sim/UI) |  | Shell | Sidebar navigation + collapse | ■ |  | Map | Basemap + overlays | ■ |  | Map | Vehicle marker + tracking | ■ (sim) |  | Map | Zoom rail | ■ |  | Map | Map HUD overlay | ■ (sim/UI) |  | Map | Home marker | ■ (UI) |  | Map | Flight trail | ■ (sim/UI) |  | Map | Manual control panel | ■ (UI) |  | Map | RTH feedback widget | ■ (sim/UI) |  | Map | Mission progress widget | ■ (sim/UI) |  | Mission | Waypoint visuals + route | ■ (demo) |  | Mission | Start gating (NFZ/prearm) | ■ (demo) |  | Mission | Mission upload/download | ■ |  | Telemetry | Right panel + live updates | ■ (sim) |  | Telemetry | Battery details / GPS quality / health | ■ (sim/UI) |  | Dock | Toolbar + logs + graph stub | ■ (demo) |  | Dock | KILL / EMERGENCY STOP UI | ■ (UI) |  |  | Commands | Arm/disarm/mode | ■ (UI/demo only) |  |  | Calibration |  | Sensors/radio/params | ■ |  |  | Connectivity | MAVLink transport | ■ |  |
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Legend:

- ■ Working properly (in current implementation)
- ■ Present but demo/placeholder (not real vehicle-backed)
- ■ Not implemented

## 9) Conclusion

The GCS you created is a **functional, polished UI shell** with **working layout behaviors, navigation, map rendering, and simulation-driven telemetry**. It is not yet a real operational GCS because it lacks the **MAVLink connectivity layer, calibration workflows, and mission protocol integration** needed to communicate with PX4/ArduPilot flight controllers.